

FHNW Rover Träff 2026

Rules 1.0

Windisch, February 1, 2026



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Versions

Version	Date	Chapter	Change
1	31.01.2026	All	First Release

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1 General Information

1.1 What is the FHNW Rover Träff?

The FHNW Rover Träff is a fascinating event that highlights top achievements in robotics! Organized by active and former members of the FHNW Rover Team, this event provides a platform for student teams from universities in Central Europe. At its core is an exciting competition where teams showcase their technical skills in designing and building a Rover each year. But the FHNW Rover Träff is more than just a competition, it is an opportunity for participants to test their rovers, exchange ideas in an informal setting, make valuable connections, and have a great time together. The various challenges not only put the technical aspects of the rovers, such as robotic arms and drive systems, to the test but also challenge teamwork.

However, this event is not only of interest to the teams themselves, as the Rover Träff also welcomes companies and organizations, offering them the opportunity to present their projects and ideas, allowing the audience to gain insight into space science and projects from companies and universities.

Immerse yourself in this exciting world of robotics, technology, and innovation - the FHNW Rover Träff is the place where ideas take off and the future is shaped!

We are happy to welcome the following teams:

- ProjectRED
- STAR Dresden e.V.
- ERIG e.V. & TUC Drillbotics & AKAMAV
- EPFL Xplore
- FRoST Frankfurt Robotics Science Team
- WARR space robotics
- Crater
- FHNW Rover Team

1.2 Schedule

1.2.1 Timeline

Date	Event
Sat. 01.12.2025	Publication of registration and information
Sun. 31.12.2025	Registration deadline
Sun. 01.02.2026	Publication of rules
March 2025	Publication of 3D models for navigation and drone task
April 2025	Publication of detailed schedule
Fri. 08 - Sun. 10.05.2026	FHNW Rover Träff

Table 1.1: General schedule

1.2.2 FHNW Rover Träff

Date	Time	Location	Program Item
Fri. 08.05.2026	17:00	Team Zone Room 1.047	Opening of Team Zone
	18:30	Team Zone Room 1.047	Team welcome, competition presentation incl. Q&A, FHNW campus tour, welcome apéro
Sat. 09.05.2026	07:00	Team Zone Room	Breakfast
	09:30	Competition area	Opening speech
	Afterwards	Competition area	Start of competition
	16:00	Competition area	End of competition
	Afterwards	Competition area	Award ceremony
	18:00	Competition area	BBQ, Stimulants and Party until open end
Sun. 10.05.2026	08:00	Team Zone Room	Breakfast
	Until 10:00	Team Zone Room	Cleanup
	Following		Optional fun program until open end

Table 1.2: Schedule FHNW Rover Träff

Spontaneous changes to Timetable due to unforeseen circumstances are possible.

1.3 Qualification and Registration

There is no qualification required for the FHNW Rover Träff. Any rover team that has been invited by the organizing committee and has registered is eligible to participate in the FHNW Rover Träff.

The participation fee is 50 CHF per person. This fee includes accommodation, meals and activities. For logistical reasons, the maximum number of participants per team is limited to 20. Of course, additional team members are welcome to support their team as fans, but may not take part in the competition.

1.4 Contacts and Information Channel

The latest information will be communicated via the FHNW Rover Träff Discord channel ([Invitation Link](#)). The main contact person for the competition is **Dominic Lehmann** (dominic.lehmann@students.fhnw.ch).

1.5 Venue and Arrival

The FHNW Rover Träff takes place at the FHNW Campus Brugg-Windisch, Switzerland. The buildings of the campus are marked in red on the map in Figure 1.1. The competition will be held in and around Building 1. The team zone is located inside building 1. Sleeping areas are provided in Building 1, showers are in building 1 (but only a few) and building 6. Facility tour will be provided on Friday.

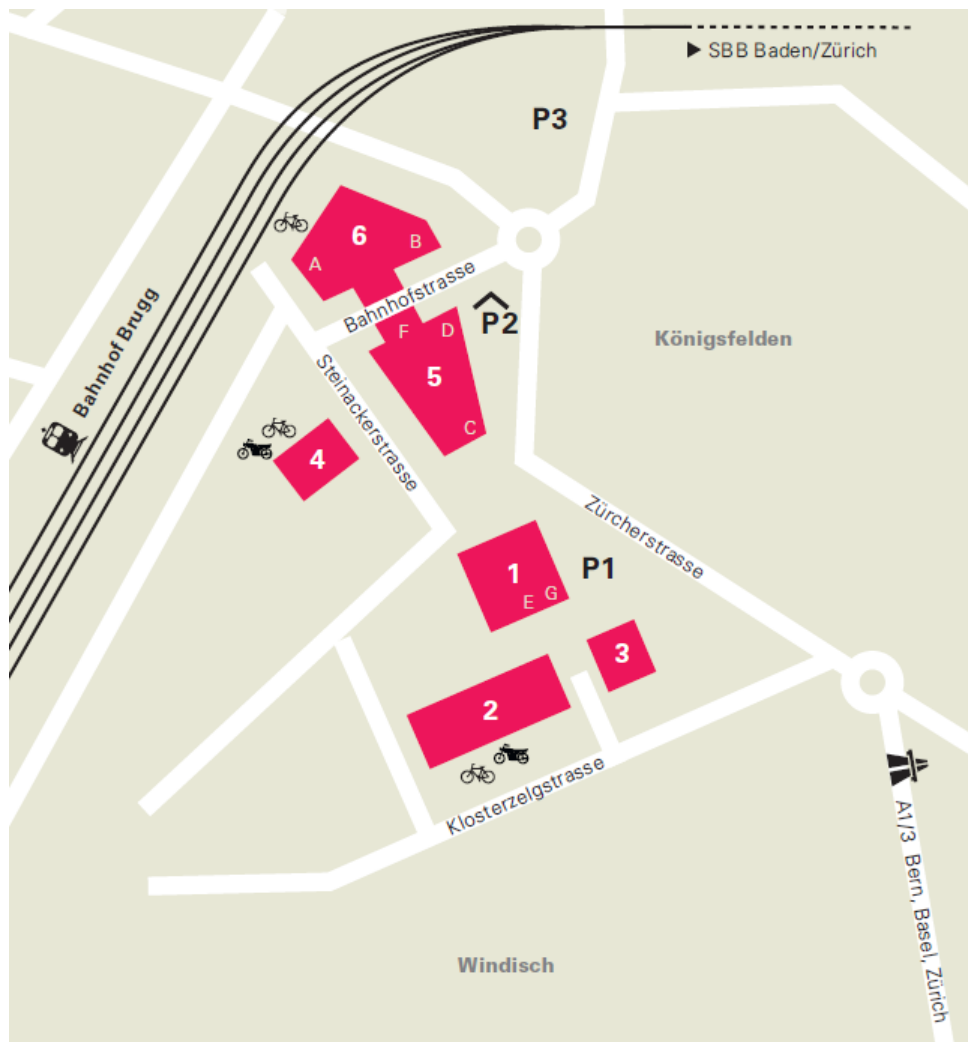


Figure 1.1: Map of FHNW Campus Brugg-Windisch

Vehicles can be parked in Parking Lot P1 ([Google Maps Link](#)). The entrance is located between Buildings 2 and 3 on Klosterzelgstrasse. If this parking lot is full, Parking Lot P3 ([Google Maps Link](#)) can be used. Each team will receive a maximum of three free exit tickets. You should drive to the parking space normally by getting a ticket. This ticket will then be exchanged for a free ticket by **Dominic Lehman** after the team welcome on Friday. More than three vehicles can be parked at your own expense.

1.6 Accommodation and Catering

The university provides sleeping rooms (lecture halls) that will be available to teams throughout the weekend. Each team has its own sleeping room, and there is one separate sleeping room for women only, which can be locked. Participants must bring their own sleeping bag and sleeping mat, since there is no furnishing to accommodate sleeping.

All drinks and food are included from Friday evening until Sunday morning. Food includes sandwiches on Friday evening, a breakfast on Saturday and Sunday morning, lunch (food truck) on Saturday and of course the BBQ after the competition.

1.7 Cost and Payment

Compared to last year, we don't want each person to pay the participation fee separately, since this caused quite an administrative overhead. Instead, the payment has to be done via bill. Each team has to provide a billing address, either University or team member which pays the entire amount for the team. It's the team's responsibility to collect the fee from each member. Each team has to provide a billing address to **Dominic Lehmann (dominic.lehmann@students.fhnw.ch)**, by end of March (31.03.2025). The billing address must contain a name and address.

2 Requirements

In addition to the informal exchange between teams, the primary focus of the FHNW Rover Träff is testing the rovers for the European Rover Challenge. For this reason, the requirements for the rover are almost identical. The organizers of the FHNW Rover Träff explicitly refrain from requiring teams to develop new systems specifically for this competition. However, it may be necessary to develop extensions for existing systems for the FHNW Rover Träff. Major extensions were communicated before the start of the fall semester so that they can be considered during the planning phase for new rovers.

The operator zone is located inside building 1, adjacent to the mars-yard. The FHNW Träff will provide Ethernet cables that are long enough to position the operator's antenna outside.

2.1 Rover

Technical Requirements:

- The rover should withstand light rain
- The battery should last at least 60 min
- The battery should be replaceable or rechargeable within 30 min
- The rover should be able to drive on dirt, grass, sand, and mud
- The rover must be equipped with an activity indicator (e.g. beacon)
- The activity indicator must be turned on when the rover is operating
- The activity indicator must be switched to a different color for autonomous operation
- The rover can use GNSS for localization
- The deep and rock sample must be stored and weighed on the rover
- The deep and rock sample must be weighed with an accuracy of 1 g
- The drilling for the deep sample must reach a depth of at least 30 cm
- All samples must be delivered across the finish line in an onboard container
- The rover must have a manipulator
- The manipulator should be able to operate a toggle switch and a push button
- The manipulator should be able to lift and hold objects with a mass of at least 500 g
- The rover should be able to connect to another 2.4 GHz Wi-Fi network during a task

Operating Requirements:

- During a mission, wired connections to the rover will result in a points penalty
- The rover must be remotely controlled via a base station
- The defined wireless communication requirements must be followed
- The rover must have an emergency stop button, which must be operable by hand
- The team shall provide a selected choice of local beer
- The drone can only be used in the drone task
- The drone task will be performed in a drone cage and therefore no legal regulations apply
- The rover must be operated without direct sight

Drone Requirements:

- The drone shall be able to be operated manually
- The drone shall be able to fly autonomously
- The drone must be able to switch from manual to autonomous mode and vice versa during flight
- The drone can use GNSS
- The drone shall be able to collect imagery of the drone cage
- A safety switch, which disables the drone, is to be implemented on the remote control
- The drone is controlled by the pilot with the remote control and one or more operators who work at the control station
- The drone shall be always in line of sight (LoS) of the pilot
- In autonomous mode, no imagery data is to be visible to the operators on the control station
- Drone must be build by the team. No COTS drones are allowed

2.2 Wireless Communication

In Switzerland, radio communications are regulated by the Federal Office of Communications (BAKOM), which has published the technical interface requirements ([RIR](#)).

These describe the conditions for frequency usage by radio equipment in the frequency range up to 3000 GHz. The RIR covers the technical parameters, frequency bands, and regulations for the use of the radio frequency spectrum. The RIR mentions the technical standards, adherence to which implies compliance with the essential requirements. Complying with these basic requirements is a prerequisite for offering, marketing, commissioning, creating, and operating radio equipment in Switzerland.

The information below serves as guidelines. The FHNW Rover Träff organizing committee (OK) assumes no liability for the use of radio equipment by the teams. The teams are responsible for complying with the radio guidelines set by BAKOM.

2.2.1 Wi-Fi

Four frequency bands are available for wireless local area networks. Teams must ensure that the maximum allowed radiated power, as defined in the interface requirements below, is not exceeded. The following summarizes the guidelines for Wi-Fi frequency bands.

2.4 GHz Frequency Band

- 2.4 GHz - 2.4385 GHz
- Indoor and Outdoor
- Max. 100 mW

The technical requirements are described in the interface requirement [RIR1010-01](#).

Lower 5 GHz Frequency Band

- 5.15 GHz to 5.35 GHz
- Indoor
- Max. 200 mW at 5.15 GHz - 5.25 GHz
- Max. 200 mW with TPC at 5.25 GHz - 5.35 GHz (DFS mandatory)

- Max. 100 mW without TPC at 5.25 GHz - 5.35 GHz (DFS mandatory)

The technical requirements are described in the interface requirement [RIR1010-05](#).

Upper 5 GHz Frequency Band

- 5.47 GHz - 5.725 GHz
- Indoor and Outdoor
- DFS mandatory
- Max. 1 W with TPC
- Max. 500 mW without TPC

The technical requirements are described in the interface requirement [RIR1010-04](#).

6 GHz Frequency Band

- 5.945 GHz - 6.425 GHz
- Max. 200 mW Indoor (LPI)
- Max. 25 mW Outdoor (VLP)

The technical requirements are described in the interface requirement [RIR1010-11](#).

2.2.2 Other Frequency Bands

For operating in other frequency bands, consult the website of [BAKOM](#) and the [RIR](#). The FHNW Rover Träff organizing committee (OK) can provide further assistance in case of any uncertainties.

2.3 Rover import into Switzerland

Teams are responsible for all required paperwork and procedure to import their equipment (Rover) legally into Switzerland from abroad. We recommend having a look at the Carnet ATA for temporarily importing foreign goods into Switzerland without national custom document. This usually includes a monetary deposit.

2.3.1 Miscellaneous

The usage of frequency bands is not limited by the FHNW Rover Träff. Teams should coordinate among themselves to ensure they do not interfere with each other during the missions. If possible the radiation should be limited outside the tasks. It's recommended to enable automatic channel selection.

3 Tasks

The competition at the FHNW Rover Träff consists of the following 5 tasks:

- Team Task
- Navigation Task
- Sampling Task
- Maintenance Task
- Drone Task

Each of these tasks will be evaluated. At the end of the competition, there will be an awards ceremony based on the overall rankings of the tasks. A detailed evaluation table will be published shortly before the FHNW Rover Träff. The jury at the FHNW Rover Träff are not members of the FHNW Rover Team and the FHNW Rover Team has the same information as guest teams. No members of the FHNW Rover Team are involved in planning of the tasks.

3.1 Background

The year is 2053. The first stable and functional research station on Ganymede, Jupiter's largest moon, is operating now for over one year without problems. The European Space Agency (ESA) and its international partners use it to further study Ganymede and plan to expand the research station's capabilities. The vision remains unchanged: to turn Ganymede into the key launchpoint for the next era of deep-space exploration.

But a sudden meteor storm has struck the research station, damaging critical external systems and scattering debris across key access routes. Fortunately, all personnel aboard the research station were unharmed. However essential infrastructure is compromised, and the station's long-term safety is at risk.

Due to severe technical issues, the crew cannot leave the station at this time. No human-led repairs, inspections, or cleanup are possible. The mission therefore depends entirely on robotics. Fortunately, rover and drone operations have continued to expand, pushing their capabilities far beyond early scouting roles and their work during the station's construction. Now they must clear hazards, assess damage, secure critical components, and bring the station back to operational status.

The mission is divided into the following five crucial tasks.

3.2 Team Task

The water pipeline was hit severely by the meteor shower, and the astronauts are left without access to drinking water. Work together with the other team to restore the pipeline and rebuild the destroyed segments. Distribute the water reserve to ensure all the settlements on Ganymed can survive!

Procedure

Preparation Time: 15min

Task Duration: 30 min

You are allowed to communicate and work together with your co-operating team in the operator zone!

The reservoir needs to be connected to three water tanks. Pipeline 1 is completely gone and needs to be rebuilt with three segments. Pipeline 2 was spared by the meteor shower and remains intact. Pipeline 3 was partially destroyed and needs one new segment.

Replace the missing pipe segments and ensure that all three empty water tanks are connected to the water reservoir. Once the pipes are connected, open the gate to fill the water tanks below. Make sure you don't waste any water, as the reservoir holds only as much water as the tanks can carry!

The task may be performed with three teams and a slightly increased work load.

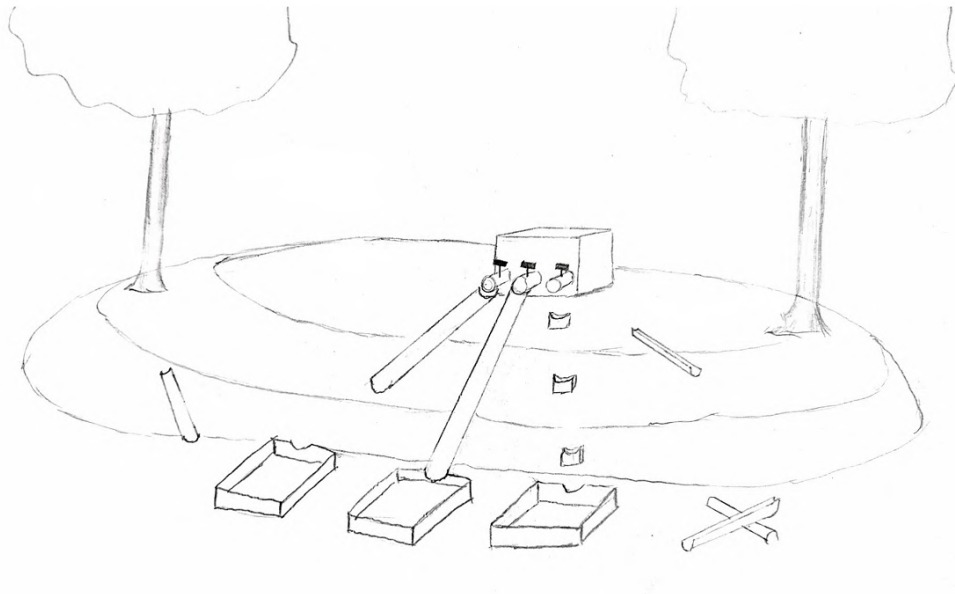


Figure 3.1: An engineers interpretation of what the task may look like

Evaluation

Team Task (150pts per team)		
Scored Parameter	max. Score	Description
Pipeline	40	10 pts per replaced pipe segment
Water Distribution	110	Water level in tanks will be measured
Penalty		-20 pts: Intervention
total	150	

3.3 Navigation Task

To ensure efficient travel between operation sites, the rover must autonomously map the terrain and determine the optimal navigation path. This task involves travelling through a series of given waypoints while using minimal remote intervention. Figure 3.2 shows the approximate space available for the task, as well as the expected position of the waypoints. Teams must develop reliable autonomous navigation systems to avoid hazards and reach their destination as efficiently as possible, the order in which the waypoints are reached is up to the teams.

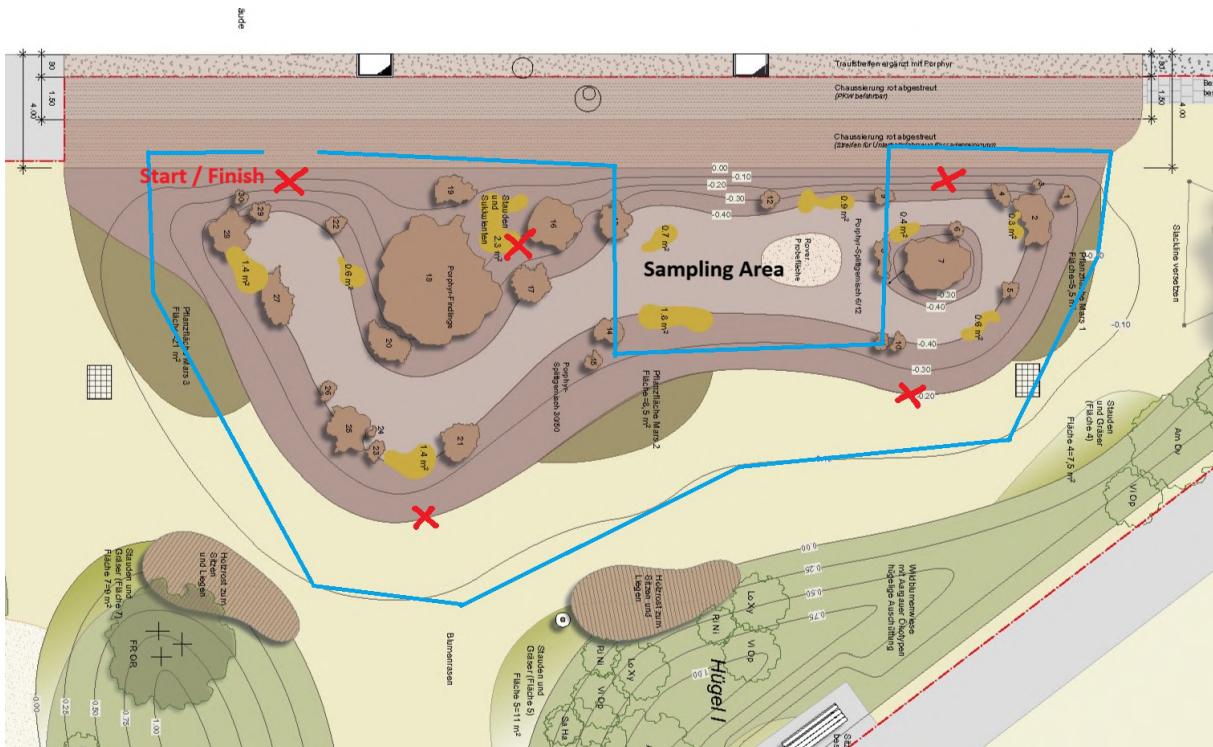


Figure 3.2: Location of the Navigation Task¹

¹Note: The Mars Yard is less “rocky” than the plan may make it seem.

Procedure

Preparation time: 15 min

Task duration: 15 min

1. Start at the predefined starting point.
2. Reach each of the four predefined waypoints one by one. After the team informs the judges that they arrived at a waypoint, one judge will measure the distance (d) to said Waypoint. The judges will give the sign to continue to the next waypoint.
3. Drive back to the starting point. This also counts as a waypoint

Evaluation

The points in the navigation task will be evaluated by reaching the four given waypoint, the finish point and the distance to each waypoint. To reach the maximum of the points the rover must drive autonomously. Points will be deducted for driving with video feed and/or GNSS. 3D point clouds do count as a video feed.

The terrains topology as well as landmarks shall be used to position the rover. The judges will penalise visual search for the waypoint indicators via camera feed.

Case Description	K_M	K_P
Waypoint reached autonomously	1	$\max\{\min\{1, 1.5 - 0.5 \cdot d\}, 0\}$
Manual driving without video feed	0.75	
Video-feed used to reach waypoint	0.5	$\max\{\min\{1, 2 - 1 \cdot d\}, 0\}$
Points per waypoint	$P_i = 30 \cdot K_M \cdot K_P$	
Penalty	-20 pts: Intervention -5 pts: Visual search for waypoints	
Points total	$\sum_{i=1}^5 P_i, P_{max} = 150$	

Table 3.1: Traverse scoring based on autonomy and distance

Landmarks

Similar landmarks will be placed on the yard as in the ERC 2025 to help for localization of the rover. The exact positions with a 3D model of the terrain will be published shortly before the FHNW Rover Träff.

Dimensions of landmarks are shown in the figure below.

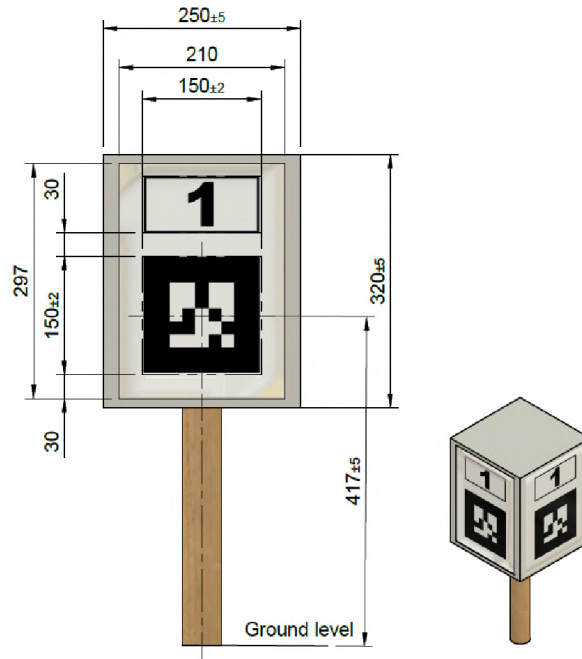


Figure 3.3: Dimensions of landmarks

Each landmark face consists of:

- A white background of at least 210 mm x 297 mm (A4 size)
- A landmark number indicator located in a 150 mm x 70 mm (± 2 mm) rectangle
- A marker square corresponding to the landmark number, with dimensions of 150 mm x 150 mm (± 2 mm) and at least 30 mm white margin on each side, grid size of the tag is 5x5

Each landmark pole consists of:

- A box with 4 landmark faces and dimensions of 250 mm x 250 mm x 320 mm (± 5 mm)
- A wooden pole that elevates the box above the ground (according to the drawing)

Each landmark's coordinates are counted from an artificial point located in the landmark's pole axis - on a height of the landmark ArUco tag centre point.

For the list of ArUco tags refer to the following [generator](#). The size 5x5 ArUco markers with the ID No. 51-64 from the generator above will be used.

3.4 Sampling Task

In order to find resources to rebuild the space station and sustain human life, the rover needs to take three different samples to evaluate where mining is suitable. This task is done in collaboration with the drone, which finds suitable and promising locations for sampling. When the rover arrives at these locations, it shall fetch three different samples: a Deep Sample, Regolith Sample and Rock Sample.

Comments

All taken samples shall be stored in a separate onboard container and brought across the finish line. The sequence in which the different samples are taken can freely be chosen. The Deep

Sample must not be taken with the manipulator, but must be taken with a specialized tool. After the task is completed, the measured weights are communicated to the judges.

Deep Sample: The rover shall extract sand from the ground to a depth of 30cm. The sample shall consist of all the layers from the ground. When extracted, the sample shall be stored in a designated container onboard the rover and brought across the finish line. The weight of the sample shall be measured / determined during the task. The Deep Sample must be taken in the designated zone defined by the judge. The sampling zone consists of colored sand, which allows to determine the depth of which the sample was taken.

Regolith Sample: The Regolith Sample shall consist of $> 100g$ of regolith (small rocks, gravel) and shall be stored in an onboard container on the rover. The Rover shall determine the exact weight of the sample to $\pm 1g$. The teams are free to choose where they want to take the Regolith Sample.

Rock Sample: The Rock Sample is a single rock, which shall have a largest dimension of $> 150mm$. The weight of the chosen rock shall be measured by the rover with a tolerance of $\pm 1g$.

Procedure

Preparation time: 15min

Time limit task: 20min

1. Start at the Start Line and arrive at the sampling area.
2. Perform Deep Sample, Regolith Sample, and Rock Sample (in any order) and store each sample on the rover.
3. Take mass measurement of each sample and take notes.
4. Cross the finish line with all taken samples on board.
5. Give weight measurements to a judge.
6. After the task, judges take and measure the weight of the Deep Sample and the Rock Sample, and check the size of the Rock Sample.

Evaluation

Sampling Task (150 pts)		
Scored Parameter	max. Score	Description
Deep Sample	50	25 pts: Sample $> 100g$ taken 25 pts: Depth $> 30cm$
Regolith Sample	25	25 pts: sample $> 100g$ taken
Rock Sample	25	25 pts: sample $> 100g$ taken
Sample Delivery	15	5 pts: per sample delivered across the finish line
Weight Measuring	35	35 pts for correct weight measuring of all three samples. -1 pt for each gram offset
Penalty		-20 pts: Intervention -50'000 pts: Deep Drill hits sewage pipe
total	150	

3.5 Maintenance Task

A critical failure has occurred in the research station's reactor system, leading to a power outage. This must be fixed immediately to ensure that life-support systems and communications remain operational.

The rovers must replace depleted fuel rods, restart the reactor, and balance the reactors power channels. Failure to do so in the specified time slot will lead to destruction of the reactor.

Procedure

Preparation time: 15 min

Task duration: 30 min

At the first panel the reactor main power lever needs to be shut OFF and after fixing the reactor, ON again. This works as follows:

1. Switch OFF Main Power Lever
2. Do tasks on second and third panel:

At the second panel, the reactor power needs to be balanced between its channels. This works as follows:

- (a) Rotational button changes power level
- (b) Press on rotational button to lock power Level and move to next channel
- (c) All power channels need to be balanced before reactor can power up

At the third panel, the reactor fuel rods need to be changed. This works as follows:

- (a) Check indicators for defective fuel rods
- (b) Remove defective fuel rods
- (c) Take replacement fuel rod which is located nearby
- (d) Place replacement fuel rod in respective holder

3. Switch ON Main Power Lever

The task is considered complete once the reactor is powered on.

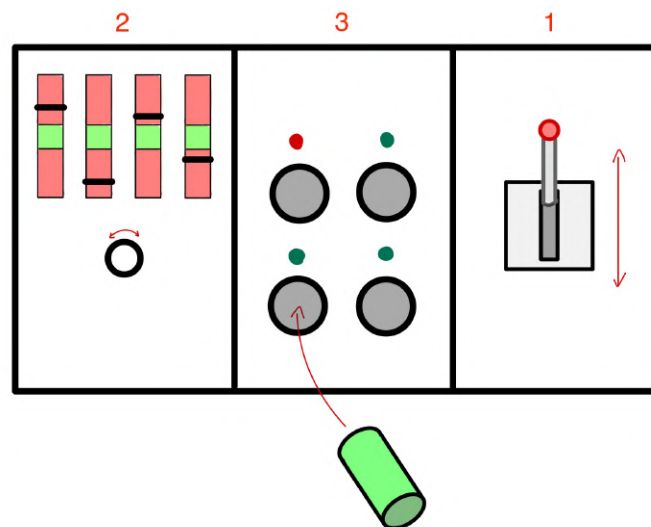


Figure 3.4: Rough sketch of panel

Evaluation

Each step performed in the procedure will earn points. The number of points depends on the complexity of the performed steps.

Maintenance Task (150 pts)		
Scored Parameter	max. Score	Description
Lever	30 pts	15 pts for turning off the power 15 pts for turning on the power
balance power channels	80 pts	20 pts per completed channel
Fuel Rods	40 pts	20 pts per fuel rod
penalty		-20 pts for intervention -20 pts for damaging the panel
total	150	

Other

The sketch is subject to minor changes. Teams will have the opportunity to inspect the panel before the task at the FHNW Rover Träff.

3.6 Drone Task

The meteor shower destroyed the storage module of the collected samples. The impact energy scattered the probes across a scree field. This field is characterized by rocks of different sizes. Since some rocks are higher than the rover, detection, maneuvering, and recovery are difficult.

Your mission: Using your drone, you conduct aerial reconnaissance of the area. The drone has to scan the entire area, detect probes and obstacles, and determine a favorable route for your rover. Recovering the probes with the rover is not part of this task, only reconnaissance, recognition and path determination are required.

Procedure

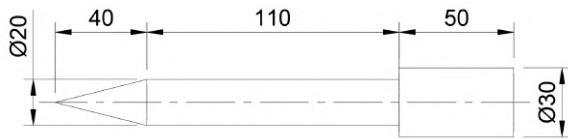
Preparation time: 15 min

Task duration: 15 min

The drone task will be performed in a drone cage with a size of 10 m x 10 m x 5 m.

1. Start at the predefined starting point
2. Scan the area using your drone. Detect, and take pictures of all objects. The objects that need to be detected are as follows:
 - (a) Probe: Metal probe identical to the probe used by ERC the previous year. The probe is made from aluminum (without surface treatment). The mechanical drawing can be seen in figure 3.5a
 - (b) Obstacles: RAKO Boxes (gray or black) with basic dimensions $600 \times 400 \text{ mm}$ and $300 \times 200 \text{ mm}$ placed bottom up, figure 3.5b
 - (c) These obstacles will be placed on a straight surface (vinyl floor)
 - (d) The number of probes and obstacles will not be communicated
3. Fly back to the starting point and land

4. Deliver a report (1 A4 page, max. 500 words) showing the detected objects, their positions relative to the starting point, and the optimal recovery route. For distance measurements, use as reference the center of gravity of each object's projected surface on the ground. For the theoretical recovery, the rover would require a circular clearance area with a diameter of 1 m to navigate between obstacles. The rover could use the whole Arena (10 m x 10 m) for maneuvering. The rover would start at the drone's starting point and must return there after recovery. No text is required, only a map including the path, obstacles and probes and a table listing each detected object, a picture of it and its position. The map can be a black white image of the arena with the contours of the objects or a few pictures cut together. However, for a judge it is difficult to believe the path was calculated using an algorithm, if the map of the arena is just a few pictures cut together. The report must be submitted as a PDF to jan.mazacek@students.fhnw.ch no later than 1 hour after completing this task.



(a) Drawing of the metal probe



(b) Example of a RAKO Box

Figure 3.5: Obstacles and probes in the arena

Evaluation

Successful completion of the drone task earns 150 points. The points will be awarded as follows:

Droning (150)		
Scored Parameter	max. Score	Description
Flight	45	15 pts: Starting 15 pts: Scanning 15 pts: Landing
Autonomy starting and Landing	15	7.5 pts: Starting 7.5 pts: Landing
Autonomy Scanning	15	autonomous scanning
Autonomy Object Detection and Photography	15	6 pts: all probes detected 1.5 pts: all probes photographed 6 pts: all obstacles detected 1.5 pts: all obstacle photographed
Autonomy Object Position Calculation	15	
Autonomy Path Calculation	15	
Report	30	6 pts: all obstacles listed with photo 6 pts: all probes listed with photos 6 pts: all obstacle positions listed 6 pts: all probes positions listed 3 pts: map of the area 3 pts: drawn route
Penalty		-20 pts: per intervention
total	150	

As long as time remains, the task may be retried as often as necessary — there is no limit on the number of attempts. The battery can be changed in between runs. However, once the time limit is reached, the drone must already have landed safely. If it is still airborne, the current attempt is disregarded.

Other

Drone autonomy is desired. The autonomy should include:

- Start and landing procedures
- Scanning of the area
- Object detection and photography
- Position calculation
- Path finding

Ten ArUco markers will be placed on the ground. The dictionary of the Markers will be 5x5 cm and the dimensions will be 15x15 cm. The marker with the ID 101 will be placed on the starting and landing location. Markers 102 -110 will be evenly distributed in the area. The position of the markers is noted in figure 3.6.

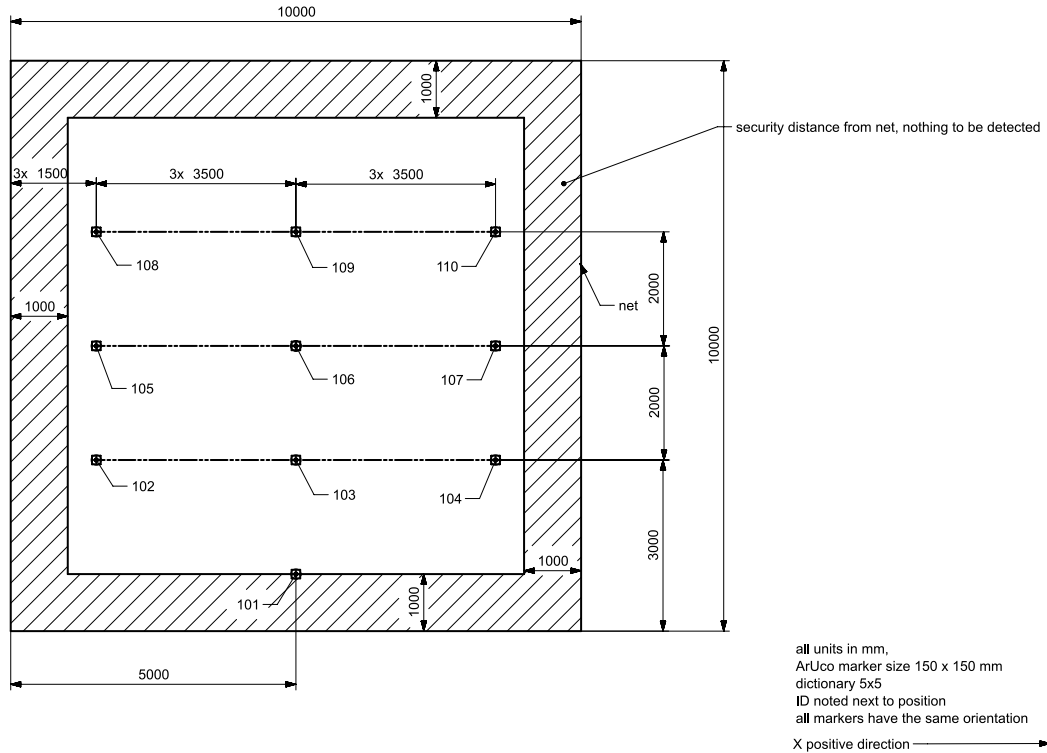
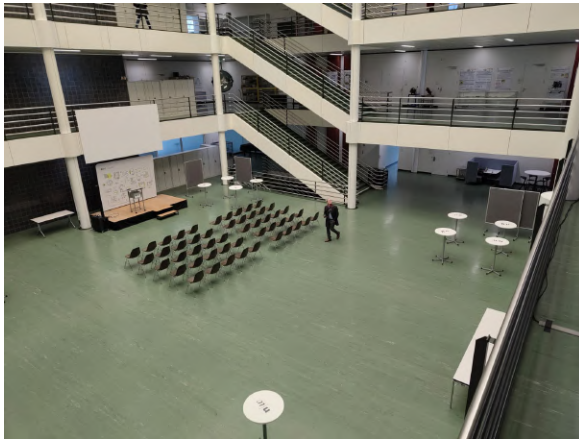


Figure 3.6: Top down view of the droning arena

The used ArUco library is <https://chev.me/arucogen/>. The necessary calculations can be done directly on the drone or on the control station which is connected wirelessly to the drone. The function of the safety switch is to be demonstrated before starting this task. During manual mode, the drone can be controlled from the remote control or the control station. The judges have to be notified before engaging autonomous mode. During autonomous mode, the pilot watches the drone and engages the safety switch if necessary. The judges shall also clearly see that the pilot doesn't influence the drone trajectory (flight control levers not engaged). No imagery data shall be visible on the control station during autonomous mode. The task will take place indoors inside Building 1 at FHNW Windisch (Lichthof), see figure 3.7a. The floor is made out of vinyl (figures 3.7c and 3.7d). Most of the light comes from light distributors of sunlight, figure 3.7e. Therefore, the lighting conditions can change, depending on the current cloudiness. The pictures of the Lichthof and the floor were taken, when there were rather few clouds. However, since it is January, the sun also is low. The cloudiness level is depicted in figure 3.7b.



(a) Lichthof at FHNW Windisch



(b) cloudiness on the day, the pictures were taken



(c) Example of the floor



(d) Example of the floor



(e) Sunlight distributors

Figure 3.7: Droning arena and its conditions

3.7 Task Schedule

The schedule for the tasks will be published shortly before the FHNW Rover Träff, by mail and over Discord. It is possible that the drone task takes place simultaneously with a rover task